Random Variables & CDFs

- Random variable (RV): real-valued function on sample space.
- Discrete RV: support finite or countable; probabilities via p.m.f. $p(x) = P(X = x)$, $p(x) \geq 0$, $\sum_x p(x) = 1$.
- Continuous RV: support uncountable; probabilities via p.d.f. $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)\,dx = 1$, and $P(a < X < b) = \int_a^b f(x)\,dx$.
- CDF (both): $F(x) = P(X \leq x)$. Discrete: $F(x) = \sum_{t \leq x} p(t)$. Continuous: $F(x) = \int_{-\infty}^{x} f(t)\,dt$ and $f(x) = F'(x)$ where differentiable.
- Key CDF properties:
  - Nondecreasing, right-continuous; limits 0 and 1 at $\pm\infty$.
  - $P(a < X \leq b) = F(b) - F(a)$.
  - Discrete point mass at b: $P(X = b) = F(b) - \lim_{x \to b^-}$ (jump height).
- Graph shapes:
  - Discrete PMF: vertical spikes at support values.
  - Discrete CDF: staircase/step function with jumps of size $P(X = x)$.
  - Continuous PDF: smooth curve (area under curve = 1).
  - Continuous CDF: smooth, non-decreasing from 0 to 1.
- From CDF to probabilities: $P(X > a) = 1 - F(a)$, $P(a < X \leq b) = F(b) - F(a)$.

Counting & Equally Likely

- Multiplication principle: m ways for step 1 and n for step 2 $\Rightarrow mn$ ordered outcomes; extend to more steps.
- Permutations (distinct): $n! = n(n-1)\cdots 1$ ways to order n distinct objects.
- Permutations with repeats: $n!/(n_1!\cdots n_r!)$, with $\sum_i n_i = n$.
- Combinations: $\binom{n}{r} = n!/[r!(n-r)!]$ ways to choose r from n (unordered).
- Multinomial coefficient: $\binom{n}{n_1,\ldots,n_r} = n!/(n_1!\cdots n_r!)$.
- Equally likely sample space: if N equally likely outcomes, $P(E) = |E|/N$.
- Typical uses: cards, lotteries, random rankings (e.g. position of top-ranked woman among 10).

Probability Laws & Bayes

- Axioms: $0 \leq P(A) \leq 1$, $P(S) = 1$, and for pairwise disjoint A_i, $P(\cup_i A_i) = \sum_i P(A_i)$.
- Basic identities: $P(A^c) = 1 - P(A)$, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.
- Conditional probability: $P(A \mid B) = P(A \cap B)/P(B)$ for $P(B) > 0$.
- Product rule: $P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$.
- Chain rule (several events): $P(A_1 \cap \cdots \cap A_n) = P(A_1) \prod_{k=2}^{n} P(A_k \mid A_1, \ldots, A_{k-1})$.
- Law of total probability: if $\{B_i\}$ partition S, then $P(A) = \sum_i P(A \mid B_i)P(B_i)$.
- Bayes' theorem: $P(B_i \mid A) = \dfrac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}$ (posterior = likelihood × prior / evidence).

Independence

- Two events A, B independent if $P(A \cap B) = P(A)P(B)$ (equivalently $P(A \mid B) = P(A)$, $P(B \mid A) = P(B)$).
- Several events $A_1, \ldots, A_n$ independent if for any subset $\{i_1, \ldots, i_k\}$, $P(A_{i_1} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^{k} P(A_{i_j})$.
- Pairwise independence does not imply mutual independence.
- Recognition: sampling with replacement $\Rightarrow$ trials usually independent; without replacement $\Rightarrow$ typically dependent (Hypergeometric).

Expectation & Variance

- Expectation: discrete $E[X] = \sum_x x\,p(x)$; continuous $E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$.
- General function: $E[g(X)] = \sum_x g(x)p(x)$ (discrete) or $E[g(X)] = \int g(x)f(x)\,dx$ (continuous).
- Linearity: $E\left[\sum_i a_i X_i + b\right] = \sum_i a_i E[X_i] + b$ (no independence required).
- Variance: $Var(X) = E[(X - \mu)^2]$ with $\mu = E[X]$.
- Variance shortcut: $Var(X) = E[X^2] - (E[X])^2$.
- Linear transform: $E[aX + b] = aE[X] + b$ and $Var(aX + b) = a^2 Var(X)$.
- Independent sum: if X, Y independent, $Var(X + Y) = Var(X) + Var(Y)$. In general, $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.
- Nonnegative RV: for $X \geq 0$, $E[X] = \int_0^{\infty} P(X > x)\,dx$.

Discrete Distributions (Formulas & Use)

- Bernoulli(p):
  - Support: $x \in \{0, 1\}$.
  - PMF: $P(X = x) = p^x(1-p)^{1-x}$.
  - Mean/Var: $E[X] = p$, $Var(X) = p(1-p)$.
  - Single success/failure trial; building block for Binomial.
- Binomial(n, p):
  - Support: $x = 0, \ldots, n$.
  - PMF: $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$.
  - Mean/Var: $E[X] = np$, $Var(X) = np(1-p)$.
  - Setting: n independent trials, success prob. p constant, RV = # successes.
- Geometric(p):
  - Support: $x = 1, 2, \ldots$.
  - PMF: $P(X = x) = (1-p)^{x-1}p$.
  - Tail: $P(X > k) = (1-p)^k$.
  - Mean/Var: $E[X] = 1/p$, $Var(X) = (1-p)/p^2$.
  - # trials until first success; memoryless: $P(X > m + n \mid X > n) = P(X > m)$.
- Negative Binomial$NB(r, p)$:
  - Support: $x = r, r+1, \ldots$.
  - PMF: $P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$.
  - Mean/Var: $E[X] = r/p$, $Var(X) = r(1-p)/p^2$.
  - # trials until r-th success; Geometric is special case $r = 1$.
- Hypergeometric(N, m, n):
  - Support: $x = 0, \ldots, n$.
  - PMF: $P(X = x) = \dfrac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$.
  - Mean: $E[X] = \dfrac{nm}{N}$.
  - Var: $Var(X) = n\dfrac{m}{N}\left(1 - \dfrac{m}{N}\right)\dfrac{N-n}{N-1}$.
  - Sampling without replacement from finite population of size N with m successes; sample size n.
- Poisson(λ):
  - Support: $x = 0, 1, 2, \ldots$.
  - PMF: $P(X = x) = e^{-\lambda}\dfrac{\lambda^x}{x!}$.
  - CDF key: $P(X \leq x) = e^{-\lambda}\sum_{t=0}^{x} \dfrac{\lambda^t}{t!}$.
  - Mean/Var: $E[X] = \lambda$, $Var(X) = \lambda$.
  - Counts of "rare" events in fixed interval; approx. $Bin(n, p)$ when n large, p small, $\lambda = np$.

Continuous Distributions (Formulas & Use)

- Uniform $U(a, b)$:
  - Support: $a < x < b$.
  - PDF: $f(x) = 1/(b-a)$ for $a < x < b$, 0 otherwise.
  - CDF: $F(x) = \dfrac{x-a}{b-a}$ for $a < x < b$.
  - Mean/Var: $E[X] = (a+b)/2$, $Var(X) = (b-a)^2/12$.
  - All values in (a, b) equally likely; $P(c < X < d) = (d-c)/(b-a)$.
- Normal $N(\mu, \sigma^2)$:
  - Support: $x \in \mathbb{R}$.
  - PDF: $f(x) = \dfrac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$.
  - CDF: $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$.
  - Mean/Var: $E[X] = \mu$, $Var(X) = \sigma^2$.
  - Standardization: $Z = (X - \mu)/\sigma \sim N(0, 1)$; use Z-table.
- Exponential $Exp(\lambda)$:
  - Support: $x \geq 0$.
  - PDF: $f(x) = \lambda e^{-\lambda x}$.
  - CDF: $F(x) = 1 - e^{-\lambda x}$, tail $P(X > x) = e^{-\lambda x}$.
  - Mean/Var: $E[X] = 1/\lambda$, $Var(X) = 1/\lambda^2$.
  - Waiting time to first event in Poisson process of rate λ; memoryless: $P(X > t + s \mid X > t) = P(X > s)$.
- Gamma $\Gamma(\alpha, \lambda)$:
  - Support: $x \geq 0$.
  - PDF: $f(x) = \dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$.
  - Mean/Var: $E[X] = \alpha/\lambda$, $Var(X) = \alpha/\lambda^2$.
  - If $\alpha = n$ integer, X is waiting time to n-th event in Poisson process with rate λ. Special case: $Exp(\lambda) = \Gamma(1, \lambda)$.

Poisson Process & Exp

- Assumptions:
  - Independent increments (disjoint intervals independent).
  - Stationary increments (rate λ constant).
  - Probability of ≥ 2 events in a very small interval is negligible.
- Rate interpretation: if events occur at rate λ per unit time, then in time t the mean count is λt and $N(t) \sim Poisson(\lambda t)$.
- Then:
  - $N(t)$, the # of events in interval of length t, has $N(t) \sim Poisson(\lambda t)$.
  - Waiting time to first event is $Exp(\lambda)$.
  - Waiting time to k-th event is $\Gamma(k, \lambda)$.

Normal, Z-scores, Quantiles

- Standardization: $Z = (X - \mu)/\sigma$.
- Forward: given x, compute $z = (x - \mu)/\sigma$, then $P(X \leq x) = P(Z \leq z)$.
- Inverse: given p, find z_p such that $P(Z \leq z_p) = p$, then $x_p = \mu + \sigma z_p$.
- Linear transformations: if $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.
- Spread comparison: larger $\sigma \Rightarrow$ wider spread, flatter peak (center given by μ).
- Empirical rule: about 68% within $\mu \pm \sigma$, 95% within

$\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$.

Normal Approx. Binomial/Poisson

- If $X \sim \text{Bin}(n, p)$ and $np(1-p) \gtrsim 10$, then $X \approx N(\mu = np, \sigma^2 = np(1-p))$.
- Continuity correction (with $Y \sim N(\mu, \sigma^2)$):
  - $\text{P(X} = \text{k)} \approx \text{P(k} - 0.5 < \text{Y} < \text{k} + 0.5)$.
  - $\text{P(X} \leq \text{k)} \approx \text{P(Y} < \text{k} + 0.5)$.
  - $\text{P(X} \leq \text{k)} \approx \text{P(Y} < \text{k} - 0.5)$.
  - $\text{P(X} \geq \text{k)} \approx \text{P(Y} > \text{k} - 0.5)$.
  - $\text{P(a} \leq \text{X} \leq \text{b)} \approx \text{P(a} - 0.5 < \text{Y} < \text{b} + 0.5)$.
- For $X \sim \text{Poisson}(\lambda)$ with large λ, $X \approx N(\lambda, \lambda)$.

Transformations $Y = g(X)$

- Discrete: if $Y = g(X)$ and X discrete, $\text{P(Y} = \text{y)} = \sum_{\text{x:g(x)=y}} \text{P(X} = \text{x)}$.
- Continuous, monotone g:
  - Step 1: find support of Y by mapping endpoints of X's support through g.
  - Step 2: find inverse $x = g^{-1}(y)$ on that support.
  - Step 3: use $\text{f}_\text{Y}(\text{y}) = \text{f}_\text{X}(\text{g}^{-1}(\text{y})) \left| \frac{\text{d}}{\text{dy}} \text{g}^{-1}(\text{y}) \right|$.
- Example: $X \sim U(a, b)$, $Y = e^X$. Then $x = \ln y$, support $e^a < y < e^b$, and $f_Y(y) = f_X(\ln y) \cdot \left| \frac{d}{dy} \ln y \right| = (1/(b-a)) \cdot (1/y)$ on (e^a, e^b).
- Example: $X \sim \text{Exp}(\lambda)$, $Y = ce^X$: same 3-step method (invert, differentiate, plug into formula).

Modeling Strategy (What to Use?)

- Binomial: "exactly k successes in n trials" (coin flips, correct answers, defective items with replacement).
- Hypergeometric: from N items, m of type 1, sample n without replacement, probability of k type 1.
- Geometric: trial of first success.
- Negative Binomial: trial of r-th success.
- Poisson: k events in fixed time/space with constant rate and independent events (typos per page, crashes per month, customers per minute).
- Exponential: time until first event when memoryless assumption is reasonable.
- Gamma: time until r-th event in Poisson process.
- Uniform: quantity equally likely anywhere in an interval.
- Normal: sums/averages of many small independent contributions; test scores, measurement errors, CLT-based approximations.

Common Approximations & Series

- Poisson approx Binomial: if $X \sim \text{Bin}(n, p)$, n large and p small, let $\lambda = np$. Then $\text{P(X} = \text{k)} \approx \text{e}^{-\lambda}\lambda^\text{k}/\text{k!}$.
- Rule of thumb: good approx if $n \geq 20$, $p \leq 0.05$, $\lambda = np$ moderate; very good if $n \geq 100$, $np \leq 10$.
- Normal approx Binomial: use when $np(1-p) \geq 10$, with continuity correction.
- Binomial theorem: $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.
- Log rules: $\log(ab) = \log a + \log b$, $\log(a^c) = c \log a$, $\log(a/b) = \log a - \log b$.